

A COMPACT CLOSED FORM FOR THE MIDDLE COMPANION ROW OF THE BROWN–ZUDILIN $\zeta(5)$ CONSTRUCTION

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ABSTRACT. Brown and Zudilin’s cellular construction for $\zeta(5)$ produces three rational sequences $Q_n, \hat{P}_n, P_n$, all annihilated by one explicit order-three operator. Only Q_n was given by a closed binomial sum; $\hat{P}_n$ and P_n are defined by the recurrence. We prove that the middle row has the two-term closed form $\hat{P}_n = \sum_{k,l} T(n, k, l) (H_{n+k}^{(3)} - \Psi H_{n+k}^{(2)})$, where T is the cellular summand and Ψ an explicit antisymmetric weight. The proof combines a Barnes–Mellin descent with Zudilin’s partial-fraction lemma, and uses no comparison of $\mathbb{Q}$ -coefficients of zeta values — a point of substance, since the required linear independence is not known. We record what the same methods do *not* give for the top row P_n , in the form of five bounded exclusions, and we state the status of an ongoing formal verification.

1. INTRODUCTION

Brown and Zudilin [1] construct, from cellular integrals on $\overline{\mathcal{M}}_{0,8}$, a family of linear forms in $1, \zeta(2)\zeta(3), \zeta(5)$ whose coefficients are three rational sequences $Q_n, \hat{P}_n, P_n$. All three satisfy a single order-three recurrence with polynomial coefficients, obtained in [1] by creative telescoping on the integral. The sequences are then *defined* by that recurrence together with initial values; the agreement of two order-three solutions at $n = 0, 1, 2$ is what proves their decomposition.

Of the three, only Q_n is given in [1] by a closed binomial sum. Our object here is the middle row.

Theorem 1.1. *For every $n \geq 0$,*

$$\hat{P}_n = \sum_{k,l=0}^n T(n, k, l) \hat{w}_3(n, k, l), \quad \hat{w}_3 = H_{n+k}^{(3)} - \Psi H_{n+k}^{(2)},$$

with T, Ψ as in Definition 2.1.

Two features of the proof are worth flagging at the outset.

First, it is not a creative-telescoping certificate. The operator L of [1] is *not* a telescoper of $T \cdot \hat{w}_3$; the minimal telescoper of that summand has order seven (§5). The proof instead runs through a Barnes–Mellin descent, where the order-three structure lives.

Second, and less obviously, the proof must avoid comparing coefficients of zeta values. The decomposition of [1] identifies the $\zeta(2)$ -coefficient of an integral with $-4\hat{P}_n$; a Barnes evaluation identifies it with something else; and concluding that the two agree requires $\mathbb{Q}$ -linear independence of $1, \zeta(2), \zeta(3), \zeta(4), \zeta(5), \zeta(2)\zeta(3)$, which is not known — indeed the irrationality of $\zeta(5)$ is open. Lemma 4.1 is what removes that step.

Status. Theorem 1.1 is proved. It is *not* machine-verified; §7 states precisely what has been formalised and what has not. The corresponding statement for the top row P_n is open, and §5 records what is excluded.

2. THE OBJECTS

Definition 2.1. Write $H_m^{(r)} = \sum_{j=1}^m j^{-r}$ and $H_m = H_m^{(1)}$. Put

$$T(n, k, l) = \binom{n+k}{n} \binom{n}{k}^2 \binom{n+l}{n} \binom{n}{l}^2 \binom{n+k+l}{n}, \quad Q_n = \sum_{k,l=0}^n T(n, k, l).$$

For $r \geq 1$ set

$$A_r(x) = H_{n+x}^{(r)} - H_x^{(r)}, \quad B_r(x) = H_{n-x}^{(r)} - H_x^{(r)}, \quad C_r = H_{n+k+l}^{(r)} - H_{k+l}^{(r)},$$

and

$$\alpha = A_1(k) - A_1(l), \quad \beta = B_1(k) - B_1(l), \quad \Psi = \frac{1}{2}\alpha + \beta, \quad L_k = -A_1(k) - C_1 - 2B_1(k),$$

with L_l the mirror of L_k under $k \leftrightarrow l$. Let L denote the order-three operator of [1], and let $\widehat{P}$ be the solution of $L\widehat{P} = 0$ with $\widehat{P}_0 = 0$, $\widehat{P}_1 = \frac{101}{4}$, $\widehat{P}_2 = \frac{344923}{96}$.

Note α, β, Ψ are antisymmetric under $k \leftrightarrow l$, while T is symmetric.

3. TWO REDUCTIONS

The first is free, and is used twice more below.

Lemma 3.1. *For any weight w , $\sum_{k,l} T w = \sum_{k,l} T w^{\text{sym}}$, where $w^{\text{sym}}(k, l) = \frac{1}{2}(w(k, l) + w(l, k))$. In particular every $k \leftrightarrow l$ -antisymmetric weight is annihilated by the double sum.*

Proof. $T(n, k, l) = T(n, l, k)$, so relabelling gives $\sum T w = \sum T w^\sigma$ with $w^\sigma(k, l) = w(l, k)$; average. $\square$

Explicitly, writing $X_r = H_{n+k}^{(r)}$ and $Y_r = H_{n+l}^{(r)}$,

$$(1) \quad \widehat{w}_3^{\text{sym}} = \frac{1}{2}(X_3 + Y_3) - \frac{1}{2}\Psi(X_2 - Y_2),$$

since Ψ is antisymmetric.

Lemma 3.2 (the $\zeta(2)$ bridge). *For every $n \geq 0$,*

$$\sum_{k,l=0}^n T \left(H_{k+l}^{(3)} + \frac{1}{4}(L_k + L_l) H_{k+l}^{(2)} \right) = \sum_{k,l=0}^n T \widehat{w}_3^{\text{sym}}.$$

Proof sketch. Both sides arise as the $\zeta(2)$ -coefficient of one Barnes-type double integral. Writing the translated rational kernel as $R(x, y) = P(x)P(y)P(x+y)/(Q(x)^2Q(y)^2)$ with $P(z) = \prod_{r=1}^n (z-r)$ and $Q(z) = \prod_{r=0}^n (z+r)$, the local data at the double pole $y = -l$ is $g_l(x) = c_l P(x)P(x-l)/Q(x)^2$, and $\sum_k B_{kl} = 0$ for each l by vanishing of the residue at infinity. Since $L_k - L_l = -2\Psi$, symmetry reduces the difference of the two sides to $-2 \sum_l \sum_{j=l+1}^n g'_l(j)$; and on $l < j \leq n$ the function g_l has a *double* zero, because $P(x)$ vanishes at $x = j$ (as $1 \leq j \leq n$) and $P(x-l)$ vanishes at $x = j$ (as $1 \leq j-l \leq n$), while $Q(j) \neq 0$ for $j > 0$. Hence every $g'_l(j) = 0$. $\square$

Remark 3.3. The vanishing ranges here are sharp: $g_l(j) = 0$ exactly on $1 \leq j \leq n+l$, $g'_l(j) = 0$ exactly on $l < j \leq n$, and the simple-pole coefficient q_l satisfies $q_l(j) = 0$ exactly on $1 \leq j \leq n$ with a zero of order exactly one.

4. THE DESCENT, AND ITS NORMALISATION

Specialising [1, (I3)] at $k = n + l$ writes the companion integral I_n'' as

$$I_n'' = \sum_l (-1)^l \binom{n+l}{n} \binom{n}{l}^2 J_3(n, n, n, n-l; n, n, n+l),$$

and the rational part of the shifted J_3 is supplied by Zudilin’s partial-fraction lemma [2, Lemma 4]: for a suitable one-variable R with $R(t) = \sum_k A_k(t+k)^{-2} + \sum_k B_k(t+k)^{-1}$, the induced quantity is $2A\zeta(3) - B$, where $A = \sum_k A_k$ and

$$B = 2 \sum_k A_k H_{k-h}^{(3)} + \sum_k B_k H_{k-h}^{(2)}.$$

For the specialisation at hand one computes, at $t = -n - 1 - k$,

$$(2) \quad A_{kl} = \binom{n+k}{n} \binom{n+k+l}{n} \binom{n}{k}^2, \quad B_{kl} = A_{kl} L_k.$$

The descent carries a normalisation, and the following lemma is what makes it harmless.

Lemma 4.1 (normalisation). *With the outer factor $(-1)^l \binom{n+l}{n} \binom{n}{l}^2$ of the specialisation,*

$$\binom{n+l}{n} \binom{n}{l}^2 \cdot A_{kl} = T(n, k, l)$$

identically in n, k, l . Consequently the $\zeta(3)$ -coefficient produced by [2, Lemma 4] equals $2Q_n$, whereas that of I_n'' is Q_n ; the descent therefore computes $2I_n''$, and the factor is 2 for every n .

Proof. The two products consist of the same six binomial factors, reordered. The second assertion follows since $Q_n \geq 1$. $\square$

Remark 4.2. This is the step that avoids any independence hypothesis. Both quantities being compared are explicitly computed *rational* coefficients inside a single derivation, not real numbers; no statement about $\mathbb{Q}$ -linear independence of zeta values is used, here or anywhere else in the proof.

Proposition 4.3. *For every $n \geq 0$, $\widehat{P}_n = \sum_{k,l=0}^n T\left(H_{k+l}^{(3)} + \frac{1}{2}L_k H_{k+l}^{(2)}\right)$.*

Proof. Combine the specialisation with (2), [2, Lemma 4] and Lemma 4.1; the outer sign cancels against the $(-1)^l$ of the $\zeta(3)$ -coefficient. $\square$

Proof of Theorem 1.1. By Lemma 3.1 the summand of Proposition 4.3 may be replaced by its symmetrisation, which is $H_{k+l}^{(3)} + \frac{1}{4}(L_k + L_l)H_{k+l}^{(2)}$. Lemma 3.2 identifies its double sum with $\sum T \widehat{w}_3^{\text{sym}}$, and Lemma 3.1 again replaces that by $\sum T \widehat{w}_3$. $\square$

5. WHAT THE SAME METHODS DO NOT GIVE

The top row remains open: we do not know a closed form for P_n . The following exclusions delimit where it is not, and are all obtained by exact computation over stated ranges or by proof, as indicated.

Proposition 5.1. *L is not a telescoper of $T \cdot \widehat{w}_3$. The minimal telescoper of that summand is $A \cdot L$ with A of order four, unique up to scalar, hence of order seven.*

The corresponding certificate is far too large to be useful: its cofactors carry $1.5 \cdot 10^6$ monomials with coefficients of 280 bits and more.

Proposition 5.2. *There is a weight w^* , explicitly a 29-monomial element of the degree-two bare alphabet, with $\sum T w^* = \hat{P}_n$ and such that L is a telescoper of $T \cdot w^*$.*

Telescoper order is thus a property of the summand, not of the sum. This is what makes Proposition 5.1 compatible with the existence of an order-three certificate.

Proposition 5.3. *No identity $T \cdot (w^* - \hat{w}_3) = \Delta_k R_0 + \Delta_l S_0$ with vanishing boundary terms exists. Any operator M with $M \cdot T \cdot (w^* - \hat{w}_3)$ a discrete divergence must annihilate Q ; consequently the minimal such order is at least four.*

Proof. For a monomial M_i maximal in the shift closure the relevant shift matrices are diagonal, so the M_i -component is a scalar equation; multiplying by T and summing over $0 \leq k, l \leq n$ telescopes the right-hand side to boundary terms, giving $d_i Q_n = 0$ where $d = w^* - \hat{w}_3$. Since $Q_n \geq 1$, $d_i = 0$; but d has 29 nonzero maximal components. The second assertion follows from the same computation applied to M , together with the fact that Q has no annihilator of order one or two with polynomial coefficients of degree ≤ 13 . $\square$

Proposition 5.4. *L is not a telescoper of $T \cdot w$ for any weight w with $\sum T w = P_n$ lying in the nine-symbol bare weight-five span of degree at most three.*

This is an exact computation: the admissible space has dimension 449 of 1270, and P_n misses its image under the sum map by 292 of 501 equations. The obstruction is sharp — no single block family excludes, but two pairs do minimally — and the same code returns the positive answer of Proposition 5.2 at weight three when run as a control.

Finally, a negative about the literature route. Zudilin’s higher-weight statement [2, Lemma 19] isolates $\zeta(5)$ for a one-variable very-well-poised R , but does not specialise to the double kernel used here; and [1] explicitly leaves the decomposition of the two-variable universal kernels undone, supplying a descent for I'' only. So the argument of §4 has, at present, no weight-five analogue.

6. THE $\zeta(3)$ ANALOGUE, WHICH IS MACHINE-VERIFIED

For contrast, and because the comparison is instructive, we record the corresponding statement one weight down, where the formalisation *is* complete.

Let $\mathcal{A}(n, k) = \binom{n}{k}^2 \binom{n+k}{k}^2$, let $a_n = \sum_k \mathcal{A}(n, k)$ be the Apéry numbers, and let b be the solution of Apéry’s recurrence $(n+2)^3 b_{n+2} = P(n+1)b_{n+1} - (n+1)^3 b_n$, $P(n) = 34n^3 + 51n^2 + 27n + 5$, with $b_0 = 0$, $b_1 = 6$.

Theorem 6.1 (machine-verified). *For all $n \geq 0$,*

$$b_n = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2 (2H_n^{(3)} - H_k^{(3)}).$$

Theorem 6.2 (machine-verified). *For all $n \geq 0$, $\sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2 (H_{n+k} + H_{n-k} - 2H_k) = 0$.*

Theorem 6.2 has a purely rational proof: with $\varphi(z) = \prod_{i=1}^n (z+i) / \prod_{j=0}^n (z-j)$, the simple poles of φ have residues $(-1)^k \binom{n}{k} \binom{n+k}{k}$ by Lagrange interpolation, and $\text{Res}_{z=m} \varphi^2 = 2\beta_m \sum_{k \neq m} \beta_k / (m-k)$ is antisymmetric in (m, k) , hence sums to zero. No contour argument and no growth estimate is required.

Both are formalised in Lean 4 with no `sorry`, and depend on only the three standard axioms (`propext`, `choice`, and `quotient soundness`). The weight $2H_n^{(3)} - H_k^{(3)}$ appears to be new; we have not located it in the literature.

The scope of Theorem 6.1 deserves the same care as §7 demands of Theorem 1.1: in the formal development b is pinned by the *recurrence* together with $b_0 = 0, b_1 = 6$, not by Apéry’s classical double sum with its nested alternating inner sum. The equivalence of the two is proved on paper by six Zeilberger/Gosper certificates and is not formalised. The situation is thus exactly parallel to the $\zeta(5)$ case, one step further advanced.

7. FORMAL VERIFICATION OF THE $\zeta(5)$ ROWS: WHAT EXISTS

Theorem 1.1 is a proof on paper. A formalisation in Lean 4 (**ZetaLucas**) is in progress and is *not* complete; we state its position precisely because the distinction is easy to lose.

Verified, with no **sorry** and only the three standard axioms: a reflective polynomial-identity checker; the summand shift calculus; the harmonic letter calculus in the shifted base; the implication “recurrence plus initial values implies the closed form”; and the initial values of the relevant double sums, computed from their definitions.

Not verified: that either double sum satisfies L . Two hypotheses remain quarantined, and the $\zeta(5)$ closed forms are consequently not machine-checked. Two further points are worth recording. First, what the order-three certificate of Proposition 5.2 would give, once complete, is $\hat{P}_n = \sum T w^*$ — *not* Theorem 1.1, since by Proposition 5.3 the two representatives cannot be bridged at order zero. Second, during this work a named hypothesis carried in the development was found to be false as stated; every theorem depending on it was **sorry**-free with clean axioms, because **#print axioms** reports what a proof depends on and not whether its hypotheses are inhabited. We mention it because the same trap is available to any development that reports conditional results.

REFERENCES

- [1] F. Brown and W. Zudilin, *On cellular rational approximations to $\zeta(5)$* , arXiv:2210.03391.
- [2] W. Zudilin, *Arithmetic of linear forms involving odd zeta values*, J. Théor. Nombres Bordeaux **16** (2004), 251–291.